

# Taeyun Roh

<https://txxnrd.github.io>

## Education

- **Korea University**, Seoul, Republic of Korea *Mar. 2022 – Present*
  - Bachelor of Science in Computer Science and Engineering

## Research Experience

- **System Software Laboratory, Korea University**  
Advisor: Professor Gyeongsik Yang *Research Intern, Jan. 2024 – Dec.2024*
  - Participated in Digital Healthcare Project **Xalute**, focusing on research and development of a wearable-based digital health platform.
  - Wrote “*Analysis on Research for Data and Resource Heterogeneity in Split-federated Learning*”, presented at the Korean Institute of Information Scientists and Engineers (KIISE) Conference 2024, and received the **3rd Place Award in the Undergraduate Category**.
  - Gave poster presentation on Secure Container for Permissioned Blockchains at KICS Summer Conference 2024.
- **Network and Mobile System Laboratory, KAIST**  
Advisor: Professor Sung-Ju Lee *Research Intern, Jan. 2025 – March.2025*
  - Focused on Efficient LLM inference including on-device LLM inference
  - Developed LLM-based methods to detect the malicious Spymic app, enhancing threat identification accuracy.
- **Data Mining Information Systems Laboratory, Korea University**  
Advisor: Professor Jae Woo Kang *Research Intern, March. 2025 –*
  - Focused on LLM knowledge and mitigating hallucination
  - Biomedical drug discovery based on LLM and schemes for mitigating hallucination

## Development & Related Experience

- **Hiconsy** *Data Analyst, Jan. 2022 – Jan. 2023*
  - Worked as a data analyst in education startup Hiconsy, now one of the biggest education companies in Korea.
  - Performed tasks like mathematical modeling for the TA salary system and regression analysis to examine the correlation between students’ study hours and academic performance.
- **KAIST MADCAMP** *Regular Participant, July 2023 – August 2023*
  - Developed game projects using Java, Swift, Kotlin, and Python.
- **Blurting** *Frontend Developer, June 2023 – Present*

- Developed a blind dating app with a completely new format. App Store link
- Ranked 73rd in App Store Social Networking Category in Korea.

- **Aigensciences**

*AI Researcher, March 2025 – Present*

- Researching LLM agent for Biomedical Domain.
- Machine Learning algorithms for Drug Discovery

## Awards

- **2nd Place in 2023 Korea University Hackathon**

- Developed a mobile application named *Noonchi* that predicts the number of visitors in Han River parks, providing real-time recommendations for less crowded parks based on user preferences.
- Check out the project on GitHub.

## Extracurricular Activities

- **Korea University Institute of Computer Security**

*Regular Member, Jan. 2023 – Dec. 2024*

- Taught C programming language to freshmen for a semester.
- Provided lectures and assignments with self-made materials on a weekly basis.
- Participated in study groups for cryptography.

- **DevKor**

*Regular Member, June 2023 – Present*

- Contributed to the frontend of the Blurting Project using Flutter.
- Enhanced knowledge of frontend technologies and cross-platform development.
- For more information, visit DevKor's page.

- **FC COMING**

*Founder and Captain, June 2023 – Present*

- Founder of a soccer club in Korea University College of Informatics.
- Participated in Korea University Sports Soccer League (KU LEAGUE) in 2023 and 2024.

- **Korea University Software Education Volunteer Group**

*Member, March 2024 – Present*  
*KUSW Facebook Page*

- Participated in software education volunteering activities, providing programming experiences and technical support to various communities.

## Projects

- **Visual Novel Game** Exciting Journey of a KAIST Student's Love Story
  - Click to Play
- **Basic CPU Implementation (COSE222 Computer-Architecture)**
  - Implemented a basic CPU architecture using SystemVerilog, featuring components such as ALU, instruction memory, data memory, controller, and register file.
  - Designed and simulated a multi-stage pipeline CPU to demonstrate efficient instruction execution.
  - View project on GitHub
- **Correlation Between Frequency Distribution Tables of KSAT and Mock Tests (2017-2021)**
  - Analyzed the relationship between the frequency distribution tables of the Korean College Scholastic Ability Test (KSAT) and mock exams from 2017 to 2021, using various regression, machine learning, and deep learning techniques.
  - View project on GitHub

## Technical Skills

- **Frontend:** HTML, CSS, React, Flutter
- **Programming Languages:** Python, C, C++, JavaScript, Dart, Rust

## Languages

- **English:** TOEIC : 940